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Iterative learning tracking control for multi-agent systems with data losses and time-correlated multiplicative noise

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Due to the time-varying fading effects of wireless channels, data losses and time-correlated multiplicative noise often occur simultaneously. This article addresses the impact of these disturbances on the consistency tracking of multi-agent systems operating repetitively. First, a transmission model is established to describe the combined influence of data losses and time-correlated multiplicative noise on consistency errors during transmission. Second, based on the established transmission model, an iterative learning control (ILC) scheme is proposed. Finally, the convergence of ILC-based multi-agent systems is analyzed. Furthermore, the analysis points out that the more sources of multiplicative noise there are, the slower the system converges. The simulation results validate the effectiveness of the proposed ILC scheme. They also confirm the analytical results regarding how the number of multiplicative noise sources affects the convergence speed of ILC-based multi-agent systems.

KEYWORDS:

Multi-agent systems, iterative learning control, time-correlated multiplicative noise, data losses, consistency error

1 | INTRODUCTION

1.1 | Backgrounds

In recent years, the research on multi-agent systems (MASs) has received widespread attention from researchers. Each agent can make independent decisions and collaborate through wireless communication, which not only enhances the system's flexibility and fault tolerance but also allows for the solution of more complex problems through collaboration. Due to the distributed working nature of the agents, such a system is easier to scale and adapt to changing environments¹. Therefore, MASs have broad application prospects in multiple fields such as intelligent transportation systems, emergency rescues, and multi-robot systems^{2,3,4}.

To achieve consistency tracking for MASs operating in repetitive tasks, iterative learning control (ILC) is an effective scheme. By utilizing previous tracking errors and appropriate learning gains to update input, ILC significantly improves the tracking performance in the iteration domain⁵. Unlike other schemes, ILC does not rely on precise object information and can effectively handle dynamic systems with high uncertainties or strong coupling. Since Arimoto proposed the ILC theory, it has achieved many research results in various aspects, especially in initial value problems, robustness, and learning algorithms^{6,7,8}.

However, the unreliability of wireless communication poses new challenges in ensuring consistency tracking of ILC-based MASs. More specifically, the input iteration for each agent depends not only on its own output error but also on the output errors of all other agents received through wireless channels. Due to multi-path and Doppler effects between transmitters and receivers,

the channel gain undergoes random fading. When the fading is shallow, it manifests as a type of multiplicative noise; when the fading is deep, the transmitted data cannot be demodulated at receivers, meaning that data losses may occur as a specific consequence of channel fading. Moreover, due to the time-varying nature of multi-path and Doppler effects, the multiplicative noise induced by channel fading is time-correlated. Therefore, it is important to investigate the impact of both data losses and time-correlated multiplicative noise on the consistency tracking of ILC-based MASs.

1.2 | Related works

To address data losses in networked ILC (NILC) systems, the Bernoulli random variable and the Markov chain are two primary modeling approaches. For instance, Liu et al. employed the Bernoulli random variable to describe data loss problems in nonlinear NILC systems⁹. Shi et al. described the random data losses in NILC systems using the Bernoulli variable¹⁰. Chi et al. introduced two independent Bernoulli random variables to express the data loss issues on the both sides of NILC systems¹¹. The data losses on both sides were also represented as a Markov chain for analyzing the convergence of NILC systems¹². Wang et al. modelled the data loss problem in switching systems using Markov chains and proposed an effective ILC method¹³. To address data loss issues in consensus control of MASs, these two modeling methods are predominantly employed as well. For example, Pan et al. used the Bernoulli variable to investigate the effects of data losses in the aircraft formation exchange process¹⁴. Shi et al. introduced the Bernoulli distribution variable to express data losses in adjacent agents¹⁵. Yan et al. introduced the Bernoulli variable to describe data losses on the actuator side in MASs¹⁶. Data losses in actuators were represented by a Markov chain to ensure the consistency of MASs¹⁷. Chen et al. used Markov chains to analyze the consistency problem in MASs under identical and non-identical data loss scenarios¹⁸. Stoica et al. characterized data losses in communication channels between adjacent agents via a Markov chain model¹⁹.

To solve the impact of data losses on the convergence of NILC systems, time-domain and iteration-domain compensations are two common methods. The former compensates for the lost data by using the received data from the previous moment in the corresponding iteration, while the latter compensates for the lost data by using the received data from the corresponding moment in the previous iteration. For example, Bu et al. employed time-domain compensation to resolve the problem of random data losses in nonlinear NILC systems²⁰. Liu et al. utilized iteration-domain compensation to solve the issue of random data losses on the input and output sides of NILC systems²¹. Huang et al. addressed data loss problems in NILC systems with continuous input by employing iteration-domain compensation²². It is worth noting that predictive compensation is the primary method to address the impact of data losses on consensus control in MASs. The basic concept of this method is to compensate for data losses by estimating the system state or output instead of merely offsetting the effects of data losses²³. For instance, Adilla et al. introduced a novel method that combines predictive strategies with gain-error-ratio-based constant control gain to address data loss issues in MASs²⁴. A prediction scheme was proposed to solve data loss issues for each agent in feedback channels²⁵. Ni et al. proposed a distributed observer based on predictive compensation to estimate the leader's state and tackle data loss problems in MASs²⁶. Additionally, a time-varying predictive method with control input quantization was proposed to actively compensate for data losses in MASs²⁷. Cao et al. employed a distributed predictive formation control protocol to reduce the influence of data losses on the consistency of MASs²⁸. Zheng et al. presented a novel predictive scheme to ensure MASs consistency in the presence of random data losses²⁹.

In addition to data losses, multiplicative noise poses significant challenges for NILC systems. Researchers have proposed various approaches to deal with multiplicative noise, which is commonly modeled as a stochastic process following the Gaussian distribution. For instance, Song et al. adopted a diagonal matrix to represent multi-channel multiplicative noise³⁰. Huang et al. designed a new input filter to guarantee the convergence of NILC systems with multiplicative noise³¹. A P-type ILC strategy was proposed to address the multiplicative noise at both output and input sides³². Bu et al. designed an event triggered model free adaptive ILC strategy to deal with the issues of multiplicative noise in systems³³. Shen et al. proposed two different learning control schemes to ensure the convergence of stochastic systems with multiplicative noise³⁴. Meanwhile, researchers investigated the impact of multiplicative noise on the consistency of MASs. For example, Lan et al. proposed a switching ILC algorithm based on observers to deal with multiplicative noise problems in MASs³⁵. Gu et al. designed a distributed sliding mode controller to solve MASs with multiplicative noise³⁶. Jia et al. designed a time-varying formation control protocol for linear MASs with multiplicative noise to ensure the consensus of leaders under different scenarios³⁷. Zhang et al. constructed a parameter-dependent Lyapunov function to address the problem of multiplicative noise in MASs³⁸. Du et al. discussed the consistency of MASs with multiplicative noise by combining event-triggered strategy and time-varying stochastic approximation gains³⁹. Zhang et al. developed a re-connection method to maintain the consistency of MASs in the presence of multiplicative

noise⁴⁰. Sun et al. proposed a distributed control protocol based on adjacent agent data sampling information to solve multiplicative noise problems⁴¹. Li et al. studied the consistency of MASs under two different multiplicative noise conditions⁴². Huang et al. proposed a reference update strategy to solve the multiplicative noise problems of MASs⁴³.

It should be noted that the multiplicative noise caused by channel fading may be not white, meaning that the multiplicative noises at adjacent times are mutually correlated. Obviously, considering the correlation of multiplicative noise is more in line with practical situations, as the multi-path and Doppler effects that cause channel fading vary over time. The research on the correlation of multiplicative noise has also received attention. For example, Liu et al. proposed an optimal filtering algorithm to ensure the stability of discrete-time linear systems⁴⁴. To further improve the system's stability, Li et al. designed a Tobit Kalman filter with censored value to solve the time-correlated multiplicative noise of the discrete-time linear system⁴⁵. Due to the great conservatism of the above literature, Wang et al. introduced new recurrence terms to estimate the system's state and time-correlated multiplicative noise, and designed a recursive estimation algorithm to resolve the noise problem of one-dimensional linear systems⁴⁶. For the increasingly complex application scenario, Gao et al. extended the time-correlated multiplicative noise problem of the one-dimensional system to the two-dimensional network control systems, and constructed a recursive estimator to ensure the system's stability⁴⁷.

1.3 | Our contribution

Some limitations in related works can be summarized as follows. First, due to the random variation of channel gain between 0 and 1, transmitted data can be disturbed by multiplicative noise caused by shallow fading or lost entirely due to deep fading, meaning data losses and multiplicative noise co-occur in wireless fading channels. However, existing research often separately addresses the problem of data losses or multiplicative noise, with little attention given to their combined influence. Second, existing work mainly studied the impact of time-correlated multiplicative noise on the system consisting of a single object and its controller, and these methods may not be directly applicable to ILC-based MASs because those interference appears in each channel among agents.

Motivated by the limitation in related works, this paper investigates the consistency tracking of ILC-based MASs under the effects of data losses and time-correlated multiplicative noise. Compared with existing works, the key contributions of this paper are as follows:

- a model is established to describe the joint influence of data losses and time-correlated multiplicative noise in the transmission process of consistency errors;
- an ILC scheme is designed to ensure the consistency tracking of MASs under the joint influence of data losses and time-correlated multiplicative noise;
- the convergence conditions of ILC-based MASs are given, and it is pointed out that the more multiplicative noise sources there are, the slower the convergence of ILC-based MASs.

The rest parts of this paper are as follows. The second section describes the research problem. The third section analyzes the convergence performance of the proposed ILC scheme. The fourth section provides simulation cases. The final section summarizes this paper.

In the paper, I_N represents an N -th order identity matrix, $\otimes$ indicates the Kronecker product, the notation $\text{diag}\{\cdot\}$ depicts a diagonal matrix, $E\{\cdot\}$ means the expectation factor.

2 | PRELIMINARIES AND PROBLEM FORMULATION

2.1 | Preliminaries

The graph theory is mainly used to describe the communication topology between multiple agents. To undirected weighted graph $G = (V, E, A)$, $V \in \{v_1, v_2, \dots, v_N\}$ indicates the nodes of the graph and each node in the set V corresponds to an agent, $E \subseteq V \times V$ denotes the edges between the nodes, and A is the adjacency matrix of the graph. In this matrix, the diagonal elements are given by $a_{m,n} = a_{n,m} > 0$, and $a_{m,n} = 0$ otherwise. $N_m = \{m \in V | (m, n) \in E\}$ is defined as the set of neighboring nodes for the m -th agent. The Laplacian matrix of the directed graph G is defined as $L = D - A$, where $D = \text{diag}(d_1, d_1, \dots, d_N)$, and $d_m = \sum_{n \in N_m} a_{m,n}$ denotes the out-degree of node m .

To ensure that MASs achieve consistency tracking of expected trajectories, a common method is to regard the desired trajectory as a virtual leader agent with the number 0, while the all agents in MASs are considered as follower agents denoted by k , where $k \in [1, 2, \dots, N]$. Subsequently, the communication topology between multiple agents can be described using a graph $\bar{G} = (v \cup \{0\}, \bar{E}, \bar{A})$ is utilized, where v represents follower agents, $\bar{E}$ indicates new boundaries, and $\bar{A}$ represents a new adjacency matrix. Moreover, the relationship between agent k and agent 0 is denoted as s_k , where $s_k > 0$ indicates a direct connection between agent k and agent 0, and $s_k = 0$ represents that the follower agent k has no connection with agent 0.

2.2 | PROBLEM FORMULATION

Consider a MASs that is composed of N homogeneous discrete-time agents, which are fully non-regular and exhibit repetitive behaviors within a finite time interval. The agent can be given as follows

$$\begin{cases} x_{k,i}(j+1) = f_k(j, x_{k,i}(j)) + Bu_{k,i}(j) \\ y_{k,i}(j) = Cx_{k,i}(j) \end{cases} \quad (1)$$

where $k = 1, 2, \dots, N$ indicates the number of agents, i denotes the iteration number, $j \in [0, T]$ represents the finite time of repetitive operation, $x_{k,i}(j) \in R^n$, $u_{k,i}(j) \in R^r$, $y_{k,i}(j) \in R^m$ respectively indicate the state, input, and output of agents, and $f_k(j, x_{k,i}(j)) \in R^{n \times n}$, $B \in R^{n \times r}$, and $C \in R^{m \times n}$ are the parameter matrices.

The desired trajectory of being seen as a virtual leader can be expressed as follows

$$\begin{cases} x_d(j+1) = f_d(j, x_d(j)) + Bu_d(j) \\ y_d(j) = Cx_d(j) \end{cases} \quad (2)$$

where $x_d(j) \in R^n$, $u_d(j) \in R^r$, and $y_d(j) \in R^m$ represent the desired state, input, and output, respectively.

Remark 1. The information exchanges between multi-agents relies on wireless channels. However, channel fading often occurs due to multi-path and Doppler effects between transmitters and receivers. It should be noted that the severity of channel fading can vary over time, appearing either shallow or deep. Thus, both forms of communication interference co-occur in wireless channels. Furthermore, due to the time-varying nature of wireless channels, multiplicative noise often exhibits time correlation, rather than being merely white.

Remark 2. To a leading-follower MASs, data losses and multiplicative noise not only occur between followers but also between the leader and followers. The reason is that only some followers can have direct access to the trajectory information from the virtual leader, the followers then relies on a communication topology to exchange information⁴⁸.

3 | ILC SCHEME DESIGN AND CONVERGENCE ANALYSIS

3.1 | ILC scheme design

The consistency error of MASs encompasses two aspects: the connection strength and the relative error. The connection strength represents the degree of connectivity between an agent and its neighboring agents in MASs, thereby determining the topology structure of the MASs. Meanwhile, the relative error refer to the output difference between intelligent agents. In this paper, s_k denotes the connection strength between the virtual leader and follower k , $a_{k,l}$ indicates the connection strength between followers k and l , $y_d(j) - y_{k,i}(j)$ is the relative output error between the virtual leader and the follower k , and $y_{k,i}(j) - y_{l,i}(j)$ represents the relative error between followers k and l . Additionally, since there may be multiple followers directly connect to the follower k , the sum of relative errors for the follower k is utilized. Therefore, the transmitted consistency error of the k -th agent at the time j in the i -th iteration is defined as follows^{49,50,51}

$$\begin{aligned} \xi_{k,i}^t(j) &= \sum_{l \in N_k} a_{k,l}(j) (y_{l,i}(j) - y_{k,i}(j)) \\ &\quad + s_k(j) (y_d(j) - y_{k,i}(j)) \end{aligned} \quad (3)$$

However, due to the shallow or deep fading of channels, consistency errors would be lost or suffered by multiplicative noise. Furthermore, the channel fading is often time-varying, meaning that the multiplicative noise is time-correlated. In this section,

the time-correlation of multiplicative noise $\beta_{k,i}(j)$ is expressed as

$$\begin{aligned}\beta_{k,i}(j) &= \delta\beta_{k,i}(j-1) + \zeta_{k,i}(j) \\ &= \sum_{t=1}^j \delta^{j-t} \zeta_{k,i}(t)\end{aligned}\quad (4)$$

where δ denotes a constant matrix, $\zeta_{k,i}(j)$ is a Gaussian distributed variable.

Using a Bernoulli distributed variable $\eta_{k,i}(j)$ to describe the effect of data losses, a transmission model can be established as follows

$$\xi_{k,i}^r(j) = \eta_{k,i}(j)\beta_{k,i}(j)\xi_{k,i}^t(j) \quad (5)$$

where $\xi_{k,i}^r(j)$ represents the consistency error received by the k -th agent through fading channels.

For the sake of exploration, we make some assumptions about wireless channels and MASs as follows:

Assumption 1. $\eta_{k,i}(j)$ is with probabilities $\text{prob}\{\eta_{k,i}(j) = 1\} = \alpha_k$, $0 < \alpha_k < 1$, where α_k is known.

Assumption 2. $\zeta_{k,i}(j)$ is with the mean $E\{\zeta_{k,i}(j)\} = p_k$, where p_k is known.

Assumption 3. The initial state for each iteration of MASs is the same, such as $x_{k,i}(0) = x_d(0)$, where $x_d(0)$ indicates the initial state of all followers.

Assumption 4. The nonlinear function $f(j, x(j))$ satisfies the global Lipschitz condition, $\|f(j, x_1(j)) - f(j, x_2(j))\| \leq b_f \|x_1(j) - x_2(j)\|$, where $b_f > 0$.

Based on the established transmission model and given assumptions, a P-type ILC scheme can be proposed as

$$\begin{aligned}u_{k,i+1}(j) &= u_{k,i}(j) + \Gamma \xi_{k,i}^r(j+1) \\ &= u_{k,i}(j) + \Gamma \eta_{k,i}(j+1)\beta_{k,i}(j+1)\xi_{k,i}^t(j+1)\end{aligned}\quad (6)$$

where $u_{k,i}(j)$ is the input for the k -th agent at the time j in the i -th iteration, Γ is the gain that needs to be selected appropriately. The convergence of this ILC scheme is analyzed in the following subsection.

3.2 | Convergence analysis

Theorem 1. For the MASs controlled by the proposed P-type ILC scheme in (6), if the following condition is satisfied, the consistency tracking of the MASs is guaranteed.

$$\|I - \alpha p \delta(j+1)(L + S) \otimes \Gamma C B\| = \rho < 1, j \in [0, T] \quad (7)$$

where $\alpha = \text{diag}\{\alpha_1^T, \alpha_2^T, \dots, \alpha_N^T\}$, $p = \text{diag}\{p_1^T, p_2^T, \dots, p_N^T\}$, and $\delta(j) = \text{diag}\left\{\sum_{t=1}^j \delta^{j-t}, \sum_{t=1}^j \delta^{j-t}, \dots, \sum_{t=1}^j \delta^{j-t}\right\}$.

Proof. To derive the convergence condition for the proposed ILC scheme, we first define

$$\begin{aligned}f(j) &= \left[f_1(j, x_{1,i}(j))^T, f_2(j, x_{2,i}(j))^T, \dots, f_N(j, x_{N,i}(j))^T \right]^T, \\ x_i(j) &= \left[x_{1,i}^T(j), x_{2,i}^T(j), \dots, x_{N,i}^T(j) \right]^T, \\ u_i(j) &= \left[u_{1,i}^T(j), u_{2,i}^T(j), \dots, u_{N,i}^T(j) \right]^T, \\ y_i(j) &= \left[y_{1,i}^T(j), y_{2,i}^T(j), \dots, y_{N,i}^T(j) \right]^T, \\ \xi_i^t(j) &= \left[\xi_{1,i}^T(j), \xi_{2,i}^T(j), \dots, \xi_{N,i}^T(j) \right]^T, \\ \xi_i^r(j) &= \left[\xi_{1,i}^T(j), \xi_{2,i}^T(j), \dots, \xi_{N,i}^T(j) \right]^T, \\ \eta_i(j) &= \text{diag}\{\eta_{1,i}(j), \eta_{2,i}(j), \dots, \eta_{N,i}(j)\}, \\ \zeta_i(j) &= \text{diag}\{\zeta_{1,i}(j), \zeta_{2,i}(j), \dots, \zeta_{N,i}(j)\}.\end{aligned}\quad (8)$$

Therefore, the MASs in (1) can be expressed in a concise form as

$$\begin{cases} x_i(j+1) = (I_n \otimes f(j)) x_i(j) + (I_n \otimes B) u_i(j) \\ y_i(j) = (I_n \otimes C) x_i(j) \end{cases} \quad (9)$$

Defining $e_{k,i}(j) = y_d(j) - y_{k,i}(j)$ as the tracking error, the consistency error in (3) can be rewritten as follows

$$\begin{aligned}\xi_{k,i}^t(j) &= \sum_{l \in N_k} a_{k,l} (y_{l,i}(j) - y_{k,i}(j)) + s_k (y_d(j) - y_{k,i}(j)) \\ &= \sum_{l \in N_k} a_{k,l} ((y_d(j) - y_{k,i}(j)) \\ &\quad - (y_d(j) - y_{l,i}(j)) + s_k (y_d(j) - y_{k,i}(j)) \\ &= \sum_{l \in N_k} a_{k,l} (e_{k,i}(j) - e_{l,i}(j)) + s_k e_{k,i}(j)\end{aligned}\quad (10)$$

Subsequently, the actual consistency error in (10) can be expressed in a concise form as

$$\xi_i^t(j) = ((L + S) \otimes I_m) e_i(j) \quad (11)$$

where $S = \text{diag}\{s_1, s_2, \dots, s_N\}$ and $e_i(j) = [e_{1,i}^T(j), e_{2,i}^T(j), \dots, e_{N,i}^T(j)]^T$.

Similarly, the received consistency error in (5) can be rewritten as

$$\xi_i^r(j) = \eta_i(j) \zeta_i(j) \delta(j) ((L + S) \otimes I_m) e_i(j) \quad (12)$$

As a result, the proposed ILC scheme in (6) can be expressed as

$$\begin{aligned}u_{i+1}(j) &= u_i(j) + (I_n \otimes \Gamma) \xi_i^r(j+1) \\ &= u_i(j) + \eta_i(j+1) \zeta_i(j+1) \delta(j+1) ((L + S) \otimes \Gamma) e_i(j+1)\end{aligned}\quad (13)$$

The tracking error $e_i(j+1)$ can be derived as

$$\begin{aligned}e_i(j+1) &= y_d(j+1) - y_i(j+1) \\ &= Cx_d(j+1) - Cx_i(j+1) \\ &= C\Delta x_i(j+1)\end{aligned}\quad (14)$$

The state error $\Delta x_i(j+1)$ can be expressed as

$$\Delta x_i(j+1) = x_d(j+1) - x_i(j+1) = [f_d(j) - f(j)] + B\Delta u_i(j) \quad (15)$$

Based on equations (14) and (15), $\Delta u_{i+1}(j)$ can be derived as

$$\begin{aligned}\Delta u_{i+1}(j) &= u_d(j) - u_{i+1}(j) \\ &= \Delta u_i(j) - \eta_i(j+1) \zeta_i(j+1) \delta(j+1) ((L + S) \otimes \Gamma) e_i(j+1) \\ &= \Delta u_i(j) - \eta_i(j+1) \zeta_i(j+1) \delta(j+1) ((L + S) \otimes \Gamma C) \Delta x_i(j+1) \\ &= b(j) \Delta u_i(j) - b_1(j) [f_d(j) - f(j)]\end{aligned}\quad (16)$$

where

$$\begin{aligned}b(j) &= I - \eta_i(j+1) \zeta_i(j+1) \delta(j+1) ((L + S) \otimes \Gamma C B), \\ b_1(j) &= \eta_i(j+1) \zeta_i(j+1) \delta(j+1) ((L + S) \otimes \Gamma C).\end{aligned}$$

Taking the norm on both sides of equation (16) and according to Assumption 4, it can be inferred that

$$\|\Delta u_{i+1}(j)\| \leq \|b(j)\| \|\Delta u_i(j)\| - b_f \|b_1(j)\| \|\Delta x_i(j)\| \quad (17)$$

Similarly, equation (15) can be expressed as

$$\|\Delta x_i(j+1)\| \leq b_f \|\Delta x_i(j)\| + B \|\Delta u_i(j)\| \quad (18)$$

According to (18), $\|\Delta x_i(j)\|$ can be derived as

$$\begin{aligned}\|\Delta x_i(j)\| &\leq B \|\Delta u_i(j-1)\| + b_f \|\Delta x_i(j-1)\| \\ &= b_f B \|\Delta u_i(j-2)\| + B \|\Delta u_i(j-1)\| + b_f^2 \|\Delta x_i(j-2)\| \\ &= \sum_{t=0}^{j-1} b_f^{j-1-t} B \|\Delta u_i(t)\| + b_f^j \|\Delta x_i(0)\|\end{aligned}\quad (19)$$

Based on equations (17) and (19), we obtain

$$\begin{aligned} \|\Delta u_{i+1}(j)\| &\leq \|b(j)\| \|\Delta u_i(j)\| + b_f \|b_1(j)\| \|\Delta x_i(j)\| \\ &= \|b(j)\| \|\Delta u_i(j)\| + b_f \|b_1(j)\| \left(\sum_{t=0}^{j-1} b_f^{j-1-t} B \|\Delta u_i(t)\| + b_f^j \|\Delta x_i(0)\| \right) \end{aligned} \quad (20)$$

Taking the λ -norm on both sides of equation (20), where $\lambda > \max[1, b_f]$, it yields that

$$\begin{aligned} \|\Delta u_{i+1}(j)\| &\leq \|b(j)\| \|\Delta u_i(j)\|_\lambda + b_f^{j+1} \|b_1(j)\| \|\Delta x_i(0)\| \\ &\quad + \frac{\|b_1(j)\|}{\lambda} \sum_{t=0}^{j-1} \left(\frac{b_f^{j-1-t}}{\lambda} \right) B \|\Delta u_i(t)\| (1/\lambda)^t \\ &= \|b(j)\| \|\Delta u_i(j)\|_\lambda + b_f^{j+1} \|b_1(j)\| \|\Delta x_i(0)\| \\ &\quad + \frac{\|b_1(j)\|}{\lambda} \sum_{t=0}^{j-1} \left(\frac{b_f^{j-1-t}}{\lambda} \right) B \sup_{0 \leq j \leq T} \{ \|\Delta u_i(t)\| (1/\lambda)^t \} \end{aligned} \quad (21)$$

Equation (21) can be expressed as

$$\begin{aligned} \|\Delta u_{i+1}(j)\| &\leq E \left\{ \left(\|b(j)\| + \frac{\|b_1(j)\| B \left(1 - \frac{b_f}{\lambda}\right)^j}{\lambda - b_f} \right) E \{ \|\Delta u_i(j)\|_\lambda \} \right. \\ &\quad \left. + E \left\{ b_f^{j+1} \|b_1(j)\| \|\Delta x_i(0)\| \right\} \right\} \end{aligned} \quad (22)$$

Taking the expectation on both sides of equation (22), we have

$$\begin{aligned} \|\Delta u_{i+1}(j)\| &\leq E \left\{ \left(\|b(j)\| + \frac{\|b_1(j)\| B \left(1 - \frac{b_f}{\lambda}\right)^j}{\lambda - b_f} \right) E \{ \|\Delta u_i(j)\|_\lambda \} \right. \\ &\quad \left. + E \left\{ b_f^{j+1} \|b_1(j)\| \|\Delta x_i(0)\| \right\} \right\} \\ &= \left(\rho + \frac{\rho_1 B \left(1 - \frac{b_f}{\lambda}\right)^j}{\lambda - b_f} \right) E \{ \|\Delta u_i(j)\|_\lambda \} + \varepsilon \end{aligned} \quad (23)$$

where

$$\begin{aligned} \varepsilon &= E \left\{ b_f^{j+1} \|b_1(j)\| \|\Delta x_i(0)\| \right\}, \\ \tilde{\rho} &= \rho + \frac{\rho_1 B \left(1 - \frac{b_f}{\lambda}\right)^j}{\lambda - b_f}, \\ \rho &= E \{ \|b(j)\| \} = \|I - \alpha p \delta(j+1)(L + S) \otimes \Gamma C B\|, \\ \rho_1 &= E \{ b_1(j) \} = \|\alpha p \delta(j+1)(L + S) \otimes \Gamma C\|. \end{aligned}$$

If $\rho < 1$ holds and a sufficiently large λ can be found, we can obtain $\tilde{\rho} < 1$. Furthermore, when $j \in [1, T]$, the input error of the controller satisfies

$$\lim_{i \rightarrow \infty} E \{ \|\Delta u_i(j)\| \} \leq \frac{\varepsilon}{1 - \tilde{\rho}}$$

Therefore, it can be shown that $\Delta u_i(j)$ converges to a bounded interval as the number of iterations increases.

Similarly, equation (19) demonstrates that $E \{ \|\Delta x_i(j)\| \}$ converges asymptotically. Furthermore, subsequent analysis based on equation (14) establishes the convergence of $E \{ \|\Delta e_i(j)\| \} = C E \{ \|\Delta x_i(j)\| \}$, the system remains convergence. $\square$

Remark 3. To ILC-based MASs with data losses and time-correlated multiplicative noise, the convergence condition is that the eigenvalue norm of the output error transfer matrices $\|I - \alpha p \delta(j+1)(L + S) \otimes \Gamma C B\|$ must be less than 1. Without multiplicative noise and data losses, the convergence condition degenerates to $\|I - (L + S) \otimes \Gamma C B\| < 1$, as derived in the ideal situation.

Remark 4. It is important to note that the previous analysis focuses solely on the multiplicative noise induced by the time-varying channel itself. However, when multiple users transmit data simultaneously on the same frequency band, each user's data

is affected not only by the channel but also by all other users. This means that the source of multiplicative noise is not unique. Therefore, when extending the analysis to scenarios with multiple sources of multiplicative noise, it can be concluded that, since the mean of each noise source lies between 0 and 1, the larger the eigenvalue norm of the output error transfer matrices, the slower the convergence of ILC-based MASs.

4 | SIMULATION RESULTS

Consider the MASs consisting of 4 brush DC permanent magnet linear motors that communicate with each other, where each motor is treated as an agent. The nonlinear representation of each motor is as follows⁵²

$$\begin{cases} \dot{x}(j) = v(j) \\ \dot{v}(j) = \frac{u(j) - (f_c + (f_s - f_c)e^{-(\dot{x}/\dot{x}_\delta)^\delta}) \text{sgn}\dot{x} - f_v \dot{x} - b_1 \sin(\omega_0 x(j))}{m} \end{cases} \quad (24)$$

where $x(j)$ indicates position (m), $u(j)$ represents thrust (N), $v(j)$ denotes velocity (m/s), m represents mass (kg), j indicates continuous time (s), f_c represents the minimum value of Coulomb friction (N), f_v denotes load parameters ($N \cdot s \cdot m^{-1}$), f_s indicates static friction (N), δ denotes empirical parameters, $\dot{x}_\delta$ represents lubrication parameters, b_1 indicates amplitude (N) and ω_0 denotes angular velocity (s^{-1}).

The model parameters are as represents $\delta = 1$, $\dot{x}_\delta = 0.1$, $u(j) = 0$, $v(j) = 0$, $m = 0.59kg$, $f_c = 10N$, $f_v = 10N \cdot s \cdot m^{-1}$, $f_s = 20N$, $b_1 = 8.5N$, and $\omega_0 = 314s^{-1}$.

The expected trajectory is regarded as the virtual leader and given as follows:

$$v_d(t) = (x_0 - x_l)(60t^3 - 30t^4 - 30t^2)$$

where $t = j/(j_l - j_0)$, x_l and x_0 represent terminal and initial position, respectively, $x_0 = 0$, $x_l = 2m$, $j_l = 1s$, the sampling interval $h = 0.01$.

Each motor in equation (24) can be discretized via the Euler method to derive a general nonlinear discrete system representation⁵³:

$$\begin{cases} x_{k,i}(j+1) = f_k(j, x_{k,i}(j)) + Bu_{k,i}(j) \\ y_{k,i}(j) = Cx_{k,i}(j) \end{cases} \quad (25)$$

where

$$x_{k,i}(j) = \begin{bmatrix} x(j) \\ v(j) \end{bmatrix}, f_k(j, x_{k,i}(j)) = \begin{bmatrix} 1 & h \\ 0 & 1 - \frac{hf_v}{m} \end{bmatrix} x_{k,i}(j) - \begin{bmatrix} 0 \\ \frac{h(b_1 \sin(\omega_0 x(j)) + f_c + (f_s - f_c)e^{-(\dot{x}/\dot{x}_\delta)^\delta}) \text{sgn}\dot{x}}{m} \end{bmatrix},$$

$$B = \begin{bmatrix} 0 \\ \frac{h}{m} \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

The communication topology structure among multi-agents is depicted in Figure 1.

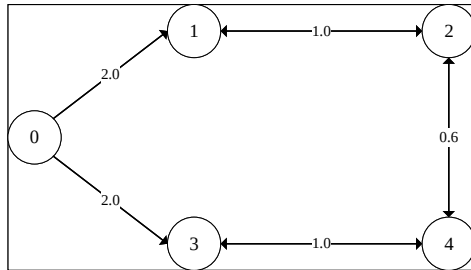


FIGURE 1 Communication topology structure among multi-agents

From Figure 1, it is clear that the agent 0 denotes the virtual leader, the agent 1 and 3 can directly connect with the virtual leader, allowing them to obtain the desired trajectory information. Agent 2 and 4 do not have direct access to the desired trajectory information, but they can communicate with agent 1 and 3 respectively. Therefore, based on graph theory, we can derive the

connection strength matrix S between the virtual leader and followers, as well as a the connection strength matrix L among the followers. They can be represented as follows

$$S = \text{diag}(2, 0, 2, 0),$$

$$L = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1.6 & 0 & -0.6 \\ 0 & 0 & 1 & -1 \\ 0 & -0.6 & -1 & 1.6 \end{bmatrix}.$$

For each communication channel among agents, the data loss rate is 0.1, the mean and the variance of $\zeta_{k,i}(j)$ are 0.8 and 0.2, respectively.

Figure 2 shows that both data losses and time-correlated multiplicative noise significantly affect the consistency error of ILC-based MASs at the 15-th iteration. However, it is clear from the output at the 60-th iteration that the proposed ILC scheme effectively ensures the consistency error of each agent. Figure 3 presents the convergence of absolute mean of each agent tracking errors. It is evident that the tracking errors of all agents converge to zero.

Using the absolute mean of agent output errors as the metric, Figure 4 represents the convergence of MASs using the proposed scheme or the scheme in 52. It is evident that the scheme designed in this paper ensures the convergence of each agent, whereas the scheme in 52 fails to achieve this. This is because the control scheme in 52, designed to mitigate the effects of fading channels, only addresses data losses in fading channels and ignores the time-correlated multiplicative noise that frequently accompanies them. Figure 5 illustrates the convergence of MASs using the proposed scheme with different number of multiplicative noise sources. It is evident that this scheme ensures the convergence of the mean consistency error regardless of whether data losses occur with 1 or 2 multiplicative noise sources. This figure suggests again that the designed scheme exhibits the usefulness for MASs implementing the proposed ILC scheme with data losses and time-correlated multiplicative noise. Furthermore, this figure indicates that an increase in the number of multiplicative noise sources reduces the convergence speed of ILC-based MASs, which verifies the correctness of theoretical analysis in Remark 4.

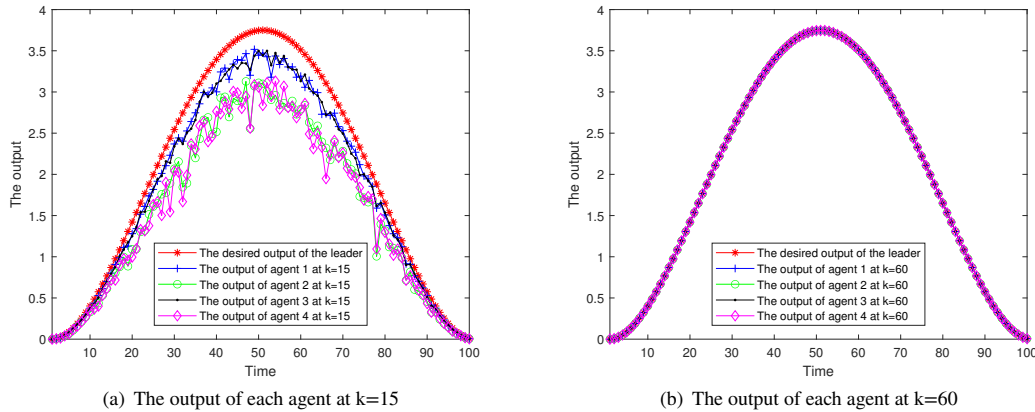


FIGURE 2 The output of each agent at different iterations

5 | CONCLUSION

This article has studied the consistency tracking of ILC-based MASs under the combined influence of data losses and time-correlated multiplicative noise. Firstly, a transmission model has been established to describe the combined influence of data losses and time-correlated multiplicative noise on consistency errors. Using the established transmission model, a new ILC scheme has been proposed to ensure the consistency tracking of MASs. Furthermore, theoretical analysis have shown the

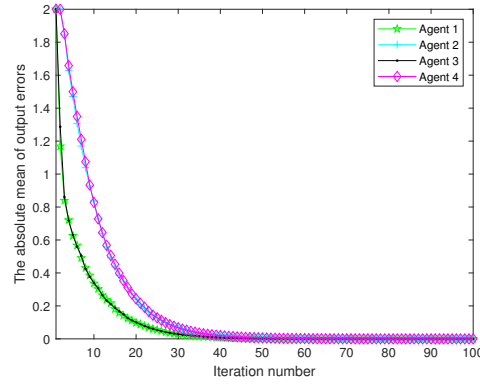


FIGURE 3 The convergence of absolute mean of each agent output errors

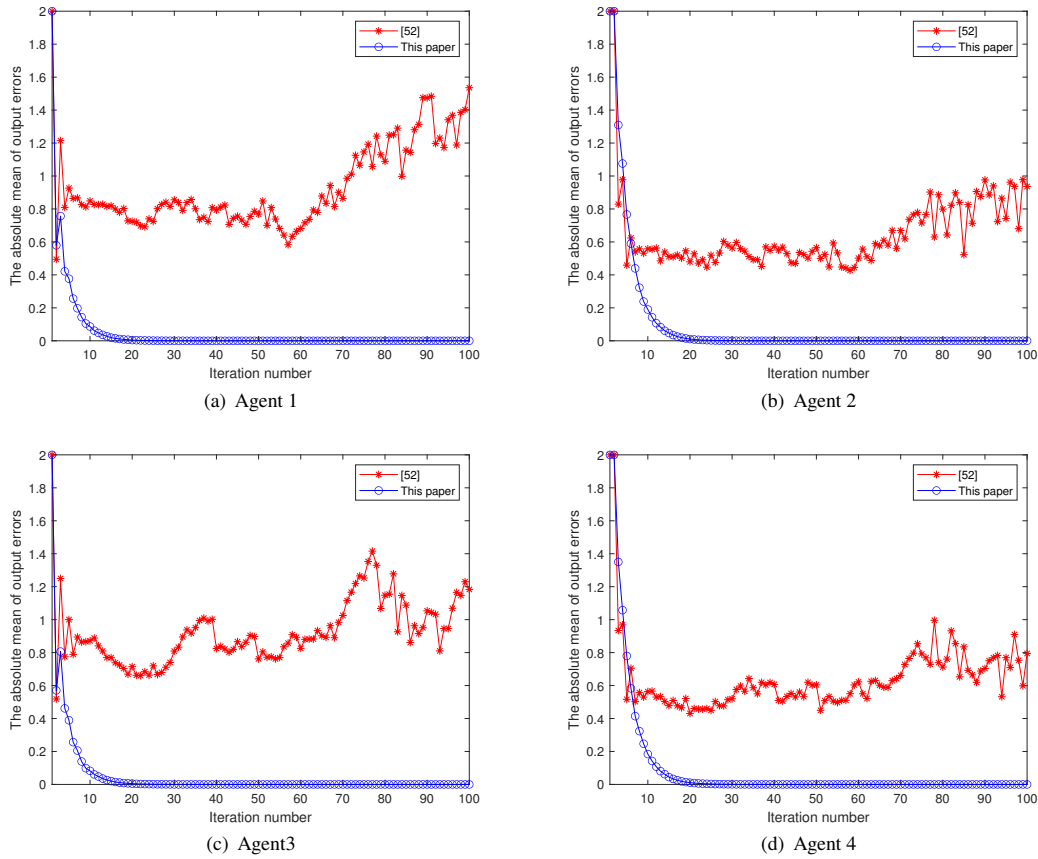


FIGURE 4 The convergence of MASs using the proposed scheme or the scheme in 52

relationship between the number of multiplicative noise sources and the convergence speed of ILC-based MASs. Finally, the effectiveness of proposed scheme has been demonstrated through experimental verification.

In future work, we will further investigate the impact of multiple multiplicative noise on the consistency tracking of MASs. Additionally, we aim to further improve the convergence of ILC-based MASs by dynamically switching to select the best topology among agents.

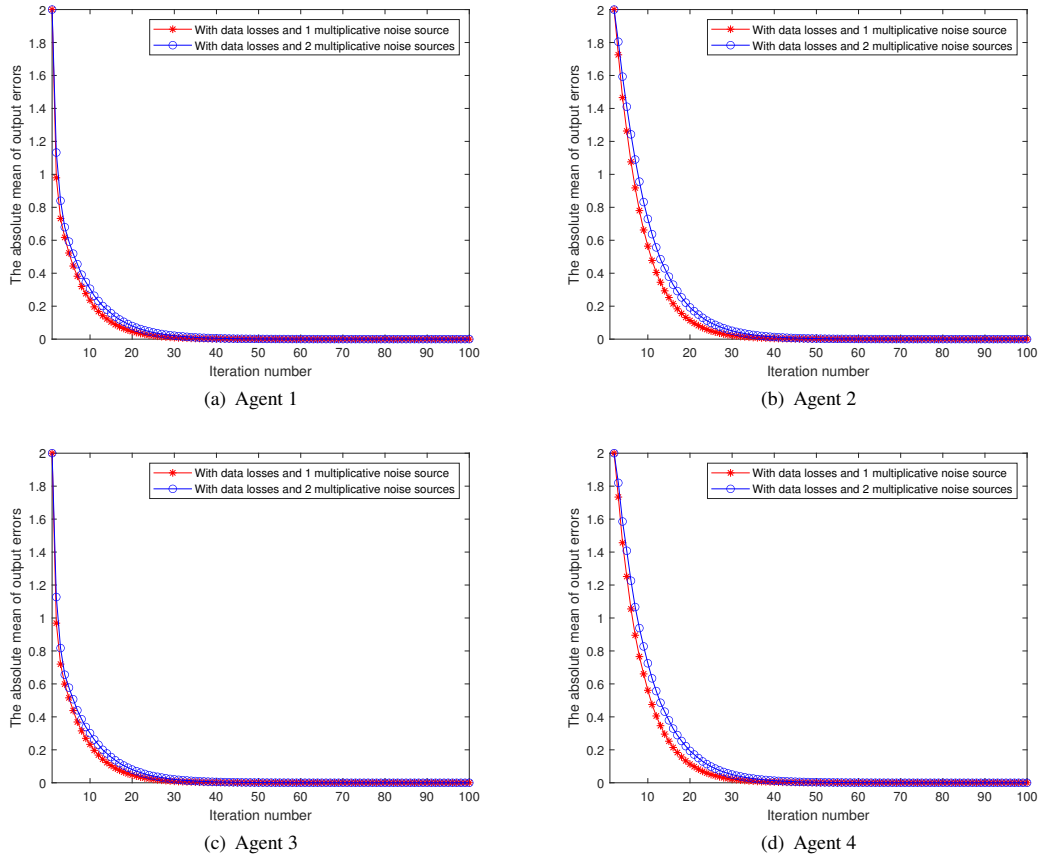


FIGURE 5 The convergence of MASs using the proposed scheme in different scenarios

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AUTHOR CONTRIBUTIONS

Lixun Huang: Conceptualization (lead); Methodology (supporting); Writing-original draft (equal); Writing-review and editing (equal). Huan Wang: Conceptualization (supporting); Methodology (lead); Writing-original draft (equal). Lijun Sun: Conceptualization (supporting); Writing-review and editing (equal). Xueyan Chen: Conceptualization (supporting). Hui Dun: Conceptualization (supporting). Qiuwen Zhang: Methodology (supporting).

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

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